

8 Memory Management

Linux was designed in the days when RAM was expensive, and it still is as compared to hard disk drive. To use a portion of your hard disk drive as a piece of RAM, Linux supports Virtual Memory. Linux kernel will keep looking for currently unused block of RAM and will transport them to the Virtual Memory to keep RAM free and available for the task at hand. To get speed advantages even with the hard disk drive, it is always recommended to maintain a separate partition for swap.

The swap file is just like any other file on your system as far as the kernel is concerned with the only restriction that it must be available locally and not mounted over the NFS.

You should also know that Linux allows one to use several swap partitions and/or swap files at the same time. This means that if you only occasionally need an unusual amount of swap space, you can set up an extra swap file at such times, instead of keeping the whole amount allocated all the time.

Just to be clear swapping and paging are almost used interchangeably as terms but the difference is that swapping involves writing the whole process out, while paging indicates towards writing only specific blocks in and out.

Creating a swap file is a simple two step process, one is creating space for it and the second being to indicate it as swap.

The first part can be done as follows:

```
$ dd if=/dev/zero of=/digit-swap bs=1024
    count=1024
    1024+0 records in
    1024+0 records out
```

1024 above is the size in kilobytes. Digit-swap is of course the name of the swap file, the point to note here is that the size should always be a multiple of 4, as the kernel writes out memory pages which are 4 kilobytes in size, so this ensures maximum efficiency. Now to mark it as swap, we use the command **mkswap**, the procedure is illustrated below:

```
$ mkswap /digit-swap 1024
    Setting up swap space, size = 1044480
    bytes
```

At this point it is critical to note that we have created the swap space and is ready to be used but the kernel will still not use it, as it is still not instructed to do so, this is analogous to having a pen drive with you but not inserting it into the USB drive. To use the file as swap, you should instruct the kernel explicitly using the command **swapon**, an example below